Science Innovation Challenge and Exhibition: Exploring Scientific Innovations

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Introduction

The **maiden edition** of the Science Innovation Challenge and Exhibition (SICE) marked a significant milestone in promoting scientific inquiry, technological advancement and entrepreneurial spirit among students. Held on July 18, 2025, this inaugural event, spearheaded by the dedicated efforts of the **Science and Innovation Committee,** brought together bright minds to present their innovative solutions to pressing global challenges, all framed within the ambitious framework of the United Nations Sustainable Development Goals (SDGs). The exhibition served as a vibrant platform for young innovators to transform theoretical knowledge into practical applications, demonstrating the immense potential within the academic community to drive meaningful change.

Out of a total of eighteen compelling submissions, nine outstanding projects were on display. These projects were not merely academic exercises but tangible efforts to address real-world problems, ranging from enhancing educational accessibility to fostering economic growth and improving public health. The diversity of the projects underscored the interdisciplinary nature of modern problem-solving and the commitment of the participants to contributing to a more sustainable and equitable world.

Featured Projects and Their Contributions

The nine projects prominently featured at the exhibition stood out for their ingenuity, relevance and potential for impact. Each project meticulously outlined its objectives, methodology, and direct contributions to specific SDGs, illustrating a holistic approach to innovation.

- **AdaptIQ:** This AI-powered learning platform is a testament to the transformative power of technology in education. Designed to personalize the learning journey for every student, AdaptIQ utilizes advanced AI (including Google Gemini) and learning science principles to create custom study plans tailored to individual goals, energy levels, and learning styles. It offers smart scheduling, interactive AI tutoring, dynamic quizzes, and curated resources, all integrated into a seamless platform. By improving educational outcomes and reducing learning inequalities, AdaptIQ directly supports **SDG 4 (Quality Education)**. Furthermore, by making AI-powered education affordable and accessible, it contributes to **SDG 10 (Reduced Inequalities)**, particularly in diverse backgrounds across Ghana and Africa. In the long term, by equipping students with essential skills, it also supports **SDG 8 (Decent Work and Economic Growth)**.
- **Corpland:** A dynamic marketplace application, Corpland is revolutionizing commerce by providing a streamlined platform for individuals and small businesses to buy and sell goods and services. Its core mission is to empower entrepreneurs to launch and scale their ventures with minimal effort. By simplifying the process of posting advertisements and connecting sellers with customers, Corpland significantly contributes to local economic development and job creation, thereby supporting **SDG 8 (Decent Work and Economic Growth)**. The

project has already demonstrated impressive traction, boasting over 1,300 monthly active users and consistent daily growth, indicating its strong market fit and potential for widespread adoption.

- **Synthesis Of Mannich Base Derivatives and Exploring Their Biological Activities (Drugs):** This critical project delves into pharmaceutical innovation, focusing on the total synthesis of Mannich-base derivatives. These compounds are being explored as potential new drugs to complement existing treatments and combat the growing challenge of antimicrobial, anthelmintic, antiviral, and anti-tumor resistance. The project also aims to enhance the solubility, bioavailability, and pharmacological potency of existing drugs while minimizing adverse effects. By developing more effective and safer medications, this initiative directly contributes to **SDG 3 (Good Health and Well-being)**, particularly Target 3.3, which aims to combat communicable diseases.
- **Feeka Instant Tomato Stew Powder (Erica):** This project, which secured first place, introduces a natural, all-in-one stew blend designed for quick and easy preparation. Inspired by the challenges students face in preparing nutritious meals, Feeka offers a healthier alternative to processed fast foods, which often contain artificial additives and high sodium levels. By using only natural, organic ingredients and herbs, and employing careful preservation techniques, Feeka provides a nutrient-dense, convenient, and affordable solution. The project significantly supports **SDG 2 (Zero Hunger)** by reducing post-harvest losses through rapid processing of fresh farm produce. It also aligns with **SDG 3 (Good Health and Well-being)** by promoting healthier eating habits, **SDG 12 (Responsible Consumption and Production)** through sustainable methods, **SDG 9 (Industry, Innovation, and Infrastructure)** by encouraging small-scale food processing, and **SDG 8 (Decent Work and Economic Growth)** by creating job opportunities. Indirectly, it addresses **SDG 13 (Climate Action)** through environmentally friendly practices.
- **Ghana Asaase:** This groundbreaking blockchain-based initiative aims to bring transparency and efficiency to land ownership management. By decentralizing land records, Ghana Asaase seeks to significantly reduce land disputes and mitigate corruption within the land sector. The project leverages smart contracts (Solidity) and integrates with the KRNL SDK, with a user-friendly React.js frontend that connects to Web3 wallets like MetaMask. The inherent transparency, security, and immutability of blockchain technology are central to this solution. By providing access to justice for all and building effective, accountable, and inclusive institutions, Ghana Asaase directly promotes **SDG 16 (Peace, Justice, and Strong Institutions)**.
- **Healthline:** A smart healthcare platform designed to improve access to quality medical support, Healthline harnesses the power of AI for symptom analysis, providing instant diagnostic suggestions and recommended next steps. The platform seamlessly connects patients with nearby health centers, pharmacies, and verified healthcare providers. Integrated messaging, video, and audio calling features enable convenient remote consultations. Crucially, Healthline incorporates a digital wallet for saving towards health needs and making payments via QR codes and merchant IDs, enhancing affordability and financial inclusion. By combining AI diagnostics, telemedicine, geolocation, and secure digital payments, Healthline significantly enhances early diagnosis, treatment accessibility, and

continuity of care, directly supporting **SDG 3 (Good Health and Well-being)**, particularly Target 3.8 on achieving universal health coverage.

- **Hostelhubb:** Securing second place, Hostelhubb is an innovative digital platform addressing critical student housing challenges, particularly in university environments. It offers a seamless solution for students to find and book suitable hostels, store their belongings safely during vacations, and even earn income through various support roles. The platform aims to alleviate the stress of accommodation management and vacation storage while simultaneously creating economic empowerment opportunities for students. Hostelhubb fosters a strong community by partnering with hostel managers to ensure verified listings and reliable services. This comprehensive student ecosystem aligns with **SDG 4 (Quality Education)**, **SDG 8 (Decent Work and Economic Growth)**, **SDG 9 (Industry, Innovation, and Infrastructure)**, **SDG 11 (Sustainable Cities and Communities)**, and **SDG 12 (Responsible Consumption and Production)**.
- **Inspectre AI:** This AI-powered surveillance platform represents a significant leap in security technology. Inspectre AI transforms traditional security cameras into intelligent, real-time observers by integrating computer vision and natural language processing. Users can interact with the system by asking questions about events (e.g., "Did anyone enter the office after 10 PM?") or setting smart triggers. The system detects faces, actions, and objects across multiple camera feeds, delivering instant, contextual insights. Designed for homes, businesses, and enterprises, Inspectre AI eliminates the need for manual footage review, making video searchable and actionable. By enabling intelligent surveillance for safer neighborhoods, public spaces, and infrastructure, Inspectre AI directly contributes to **SDG 11 (Sustainable Cities and Communities)**, specifically Target 11.7 on providing safe and inclusive public spaces.
- **Lenz:** An agritech initiative, Lenz is transforming Ghanaian farming through the strategic application of AI-powered drones and intelligent data analytics. Developed over three years of research and testing, Lenz targets key agricultural challenges such as inefficient agrochemical use, delayed disease detection, labor shortages, and inaccurate yield forecasts. The drones utilize advanced sensors, AI, and real-time imaging to deliver precision spraying, early disease alerts, and data-driven decision support. The overarching goal is to empower small to medium-scale farmers with efficient and sustainable tools for improved productivity, laying the foundation for smarter agriculture across Africa. This project significantly contributes to **SDG 1 (No Poverty)**, **SDG 2 (Zero Hunger)**, **SDG 3 (Good Health and Well-being)**, **SDG 8 (Decent Work and Economic Growth)**, and **SDG 13 (Climate Action)**.

Exhibition Results

The judging panel, comprising lecturers from various fields, meticulously evaluated each project based on criteria such as innovation, feasibility, impact and alignment with the SDGs. After careful deliberation, the top three projects were announced, celebrating their exceptional contributions and awarding them well-deserved cash prizes **provided by Professor Leonard Kofitse Amekudzi, Provost of the College**:

1. **Feeka Instant Tomato Stew Powder (Erica):** This project's focus on a healthier, convenient food solution with strong implications for food security and public health resonated strongly with the judges, earning it the top honor and a cash prize of **GHS 5,000**.
2. **Hostelhubb:** Recognized for its comprehensive approach to student welfare, addressing accommodation, storage, and economic opportunities, Hostelhubb secured a well-deserved second place and a cash prize of **GHS 3,000**.
3. **Synthesis Of Mannich Base Derivatives and Exploring Their Biological Activities (Drugs):** The critical research into developing new and more effective pharmaceutical solutions for widespread health issues earned this project the third position and a cash prize of **GHS 2,000**.

Conclusion

The maiden Science Innovation Challenge and Exhibition was a resounding success, serving as a powerful testament to the ingenuity and dedication of the participating students. The projects showcased not only cutting-edge scientific and technological advancements but also a profound understanding of societal needs and a commitment to sustainable development.

This event has laid a strong foundation for future SICE editions, promising to continue nurturing young talent and fostering a vibrant culture of innovation. By providing students with a platform to transform their ideas into tangible solutions, the SICE plays a vital role in preparing the next generation of leaders and problem-solvers who will drive Ghana and Africa towards a more prosperous and sustainable future. The enthusiasm and quality of the projects presented indicate a bright outlook for innovation, demonstrating that with the right support and platforms, students can indeed be at the forefront of addressing global challenges.